

Digital Audio Player Project Brief Rev 6

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Attribution

- **Author:** Alexander Loker. Project concept, design decisions, component selection, analysis, and verification.
- **Drafter/documenter:** Claude, writing from Alexander's direction and source data.

A portable digital audio player designed from scratch by Alexander Loker: schematic, PCB, simulation, firmware, and enclosure. The device is intended as a genuine end product (small, pleasant to use, and measurably good), and the project also closes two deliberate skill gaps: custom MCU board design, and analog and mixed-signal design backed by simulation.

Changes from Rev 5:

- **Clocking:** the ES9038Q2M runs as I2S master with its DPLL off (synchronous mode); the ESP32-S3 is I2S slave. Oscillators selected: ECS-2520MVLC-225.792-CN-TR (22.5792 MHz) and Abracon ASTX-H11-24.576MHZ-T (24.576 MHz).
- **I/V stage:** OPA1612 selected. Op-amp + inputs at 0 V, $R_f = 499\ \Omega$, $C_f = 1.5\ \text{nF}$ COG.
- **Grounding:** single GND net and one solid plane.
- **Removed:** the charging section, the revision roadmap (the design goes directly to a single integrated PCB), the simulation-limits section, and the signal-chain simulation tool choice.

1. Objectives

#	Objective	How it is satisfied
1	Build an audio player worth using	Portable microSD player, FLAC and MP3, single-ended 3.5 mm output tuned for IEMs, 2.4 inch display.
2	Close the MCU-board knowledge gap	Custom ESP32-S3 board from the module up: power tree, boot straps, SDMMC, USB, I2S, I2C charger control.

#	Objective	How it is satisfied
3	Work with data converters	ES9038Q2M current-output DAC with a discrete I/V and filtering chain designed from scratch.
4	Work with analog signals	I/V conversion, differential summing, reconstruction filtering, gain structure, output impedance control, low-noise power conditioning.
5	Simulate where simulation earns its place	LTspice for the buck-boost, inverter and LDOs; signal-chain simulation where it informs a decision. See Section 6.
6	Learn	Each design decision documented and verified by measurement on the finished board.

2. Locked Scope

Decision	Choice	Rationale
MCU	ESP32-S3	USB-OTG device peripheral, 4-bit SDMMC, dual-core 240 MHz, octal PSRAM. ESP-IDF / FreeRTOS.
DAC	ES9038Q2M	Differential current output, which forces the full analog chain. Runs as I2S master with the DPLL off; all audio clocks derive from a local oscillator.
Source	microSD, local files only	No Bluetooth, no streaming, no USB Audio Class.
Formats	FLAC 16/44.1 minimum, MP3	Decoded on the MCU to PCM. The DAC never sees a file format.
USB-C role	Charging + file transfer only	TinyUSB MSC exposes the microSD as a drive.

Decision	Choice	Rationale
Output	3.5 mm TRS, single-ended, DC coupled	Balanced dropped. DC coupling preserves the sub-1-ohm target but requires offset protection.
Load target	Simgot EW300 ($28\ \Omega \pm 15\%$, 119–121 dB/V); 16–32 ohm IEMs	Unforgiving of noise floor and output impedance.
Maximum output	0.6 V RMS (0.85 V peak)	114.6 dB SPL peaks on the EW300 (worst nozzle). ≈ 98 mA peak per analog rail at $23.8\ \Omega$.
I/V op-amp	TI OPA1612AIDRGR, 8-SON	Four I/V channels in two dual packages. 1.1 nV/VHz, 40 MHz GBW.
Output amp	TI OPA1622, 10-pin VSON (DRC)	Dual channel. Thermal pad connects to V–, not ground. EN pin used for shutdown.
Analog rails	± 3.3 V	Supply range is ± 2 V to ± 18 V on (V+)–(V–). Gives ~ 2.4 V headroom at the 0.85 V peak output.
Display	2.4 inch IPS, 240x320, SPI	Glass ($\sim 43 \times 60$ mm) fits inside the battery footprint. ST7789 or ILI9341. No touchscreen.
Battery	Jauch LP605060JU+PCM, 1S1P 3.7 V, 1850 mAh	63.0 x 51.0 x 6.6 mm actual envelope, integrated PCM, 70 mm flying leads, no internal NTC.
Charger	TI BQ24250RGET	3 MHz synchronous buck with PowerPath, VQFN-24 4x4 mm. I2C (0x6A) plus standalone fallback. TS pin gives hardware JEITA. Charge at 500 mA (0.27C). Active production status verified.

Decision	Choice	Rationale
Charger variant choice	'250 spin specifically	The bq24250 carries EN1/EN2 rather than D+/D-, so it does not contend for the USB data lines. The bq24251 is NRND; the bq24253 is standalone-only.
Analog pre-regulator	ADI LTC3440, SYS $\rightarrow$ 3.9 V	1.2 MHz synchronous buck-boost covering the full 3.0–4.2 V cell range. Feeds the LT3042 and the LT3462A inverter.
Negative pre-regulator	ADI LT3462AES6, +3V9 $\rightarrow$ -3.9 V	2.7 MHz fixed-frequency dual-inductor inverter, integrated Schottky, TSOT-23-6. Regulated output, so LT3093 headroom no longer depends on source resistance.
Analog LDOs	ADI LT3042 / LT3093	± 3.32 V, ultralow noise. Current limit programmed to 125 mA on both (R_ILIM 1.0 k Ω / 15.6 k Ω), capping the LTC3440 fault load at $\approx$ 250 mA. LT3093 exposed pad is IN (-3V9).
Fuel gauge	Voltage-based, no coulomb counter	Load-compensated lookup table characterised from the spare cell. Requires ESP32-S3 ADC calibration and a gated divider.
Power silicon rule	LTspice-native where simulation drives design	Applies to the buck-boost, inverter and analog LDOs. The charger is exempt: its CC/CV state machine, termination and thermal foldback are not SPICE-characterisable. TPS63020 reuse remains excluded because the buck-boost is simulation-driven.
EDA / CAD	Altium (AL-PN library), SolidWorks	Existing toolchain carries over.

3. Explicitly Out of Scope

Item	Status	Note
4.4 mm balanced output	Dropped	Would require four amp channels.
Bluetooth / A2DP	Dropped	ESP32-S3 is BLE-only.
USB Audio Class	Deferred	Full-speed USB caps UAC1 around 24-bit / 96 kHz.
Battery telemetry / coulomb counting	Rejected	Voltage-based gauging adopted. The charger provides fault and loop status over I2C but no battery-voltage ADC.
BC1.2 / USB-C PD negotiation	Rejected	Input current limit set by EN1/EN2 pins or I2C. 5.1 kohm CC pull-downs only.
Touchscreen	Rejected	Encoder and buttons are better in-pocket.
Line-in / recording	Deferred	Nothing in the design forecloses it.
DSP / parametric EQ	Deferred	Added in firmware once the signal path is trusted.
Bare-metal firmware	Rejected	Not where the learning value is.
Vented enclosure / active cooling	Rejected	Heat reaches the cell by conduction, not convection; venting would not interrupt that path and would admit dust and moisture. A fan is also an acoustic noise source in a listening device. Made moot by the switching charger.

4. System Architecture

Signal chain: microSD → SDMMC (4-bit) → ESP32-S3 (decode to PCM in a PSRAM ring buffer) → I2S → ES9038Q2M → I/V conversion → differential summing and MFB low-pass → OPA1622 output stage → 3.5 mm TRS. A local audio oscillator drives the DAC's XI pin, and the DAC generates BCLK and LRCLK for the ESP32-S3.

Power chain: Li-Po cell → BQ24250 (charge + PowerPath, SYS output) → LTC3440 buck-boost to 3.9 V → LT3042 to +3.3 V, and LT3462A inverter to -3.9 V → LT3093 to -3.3 V. The digital 3.3 V rail is a separate branch from SYS.

Grounding: a single GND net on one uninterrupted ground plane. Analog and digital sections are separated by placement and pour partitioning, never a split plane.

+3V3_ANALOG and +3V3_DIG remain separate supply rails.

4.1 Digital subsystem

- ESP32-S3 with octal PSRAM. Decode on one core, UI and I2S service on the other.
- SDMMC in 4-bit mode. Ring buffer sized against worst-case card latency, with minimum occupancy instrumented. A load switch on card VDD allows power-gating between bursts; host pins are tri-stated before gating, since power must reach VDD before any I/O pin goes high.
- FLAC and MP3 decode entirely on the MCU. Format support costs cycles and RAM only.
- UI: 2.4 inch SPI TFT, EC11 encoder, tactile switches. Partial updates only: a full 240x320 frame at 16 bpp is 153.6 kB, about 31 ms at 40 MHz SPI.
- Display off during playback with wake on interaction. A power requirement, not a preference.
- Charger control: I2C at address 0x6A, plus two GPIOs to EN1/EN2 for the input current limit and one to /CE. Charger STAT and INT are open-drain into GPIOs through 10 kohm pull-ups.

4.2 Clocking

Jitter-limited SNR is $-20 \log_{10}(2\pi f t_j)$: roughly 98 dB at 100 ps and 118 dB at 10 ps, referred to a 20 kHz signal. Periodic jitter from a fractional-N PLL is worse than its RMS figure suggests because it produces discrete sidebands rather than a raised noise floor.

- Two oscillators: 22.5792 MHz (ECS-2520MVLC-225.792-CN-TR, 44.1 kHz family) and 24.576 MHz (Abracon ASTX-H11-24.576MHZ-T, 48 kHz family). Ratio 147/160, so neither derives from the other without a fractional PLL.
- The ES9038Q2M runs as I2S master with its DPLL off (synchronous mode, MCLK = 128·FSR). It divides the active oscillator down to BCLK and LRCLK and drives them to the ESP32-S3, which runs as I2S slave. Every audio clock derives from one local oscillator, so ESP32-S3 clock jitter does not reach the conversion.

- The unused oscillator is disabled through its OE pin (tri-state output) rather than muxed through logic. Both outputs join at XI through 22 Ω series resistors, and 10 k Ω pull-downs on both OE lines hold both oscillators off until firmware enables one.
- The oscillators need a low-noise supply: supply noise becomes phase noise via the supply pushing figure.
- Oscillators sit adjacent to the DAC XI pin. They are specified by phase noise, not frequency accuracy.

4.3 Analog chain

- The ES9038Q2M presents differential current outputs: an inverting I/V stage per phase (four channels), then differential-to-single-ended summing per channel, then the output driver.
- The I/V stage is a first-order low-pass via its feedback capacitor. The summing stage is a second-order multiple-feedback (MFB) section.
- MFB over Sallen-Key. Sallen-Key's stopband degrades at high frequency as the op-amp's rising output impedance turns the feedback capacitor into a feedthrough path, limiting rejection to roughly 40–60 dB. MFB keeps rolling off and has a virtual-ground summing node with no common-mode swing.
- Target: flat to 20 kHz within 0.1 dB, –3 dB around 60–100 kHz, well attenuated by 1 MHz.
- Op-amp allocation: OPA1612 for the four I/V channels; OPA1622 for the output stage.
- I/V operating point: op-amp + inputs at 0 V, $R_f = 499 \Omega$, $C_f = 1.5 \text{ nF}$ C0G (corner $\approx 213 \text{ kHz}$, $\approx 0.04 \text{ dB}$ loss at 20 kHz). The DC current from the DAC's current-mode offset is common to both legs of a channel and cancels in the summing stage.
- C0G/NP0 throughout the signal path. Class-II ceramics are piezoelectric and voltage-nonlinear.

4.4 Output stage

$V_{\text{load}} = V_{\text{amp}} \times Z_{\text{load}} / (Z_{\text{out}} + Z_{\text{load}})$. Multi-driver IEM impedance swings with frequency, so non-negligible Z_{out} produces audible response tilt. Target under 1 ohm.

- Closed-loop output impedance is the open-loop value divided by loop gain, so it rises with frequency. It is plotted, not quoted as a single number.
- Output-inclusive compensation: audio-band feedback after the isolation resistor, high-frequency feedback before it through a small capacitor.
- DC coupling requires output offset detection and a muting device, since a DC fault destroys an IEM. The OPA1622 EN pin participates but needs a ramp to avoid pops.

- Gain structure: design for 0.6 V RMS maximum (0.85 V peak) and attenuate ahead of the output stage. Sized for the EW300: $SPL = 119 + 20 \log_{10}(0.6) = 114.6$ dB peak (116.6 dB with the 121 dB/V nozzle). Peak signal current 35.7 mA per channel into 23.8Ω (28 Ω –15%). The lower gain also lowers the noise floor into a sensitive IEM.
- Copper resistance matters at a 1 ohm budget: 1 oz, 10 mil, 1 inch is about 50 milliohm, and the ground return counts equally. Output nets are poured.

4.5 Power tree

- Topology: cell (3.0–4.2 V) → BQ24250 SYS → LTC3440 buck-boost to 3.9 V (1.2 MHz) → LT3042 to +3.3 V; LT3462A inverter to –3.9 V (2.7 MHz) → LT3093 to –3.3 V. LDO dropout sets the 3.9 V intermediate rail; the buck-boost covers the full cell range.
- The LT3042 holds high PSRR into the MHz region where switcher energy lives, unlike a conventional LDO. Roughly 0.8 μ V RMS output noise over the audio band.
- Negative rail: LT3462A inverter, not a charge pump. Charge pumps were rejected on output resistance (LM27762 2.5 Ω , LTC3260 32 Ω) against a regulated need of < 1.7 Ω ; the LTC3260 / LTC3265 also need 4.5 V minimum input. The LT3462A switch carries both inductor currents: ≈ 0.29 A at the 98 mA full-scale peak vs its 300 mA minimum limit.
- Digital and analog take separate branches from SYS so that SD, display and backlight load steps do not modulate the analog rail.
- Switchers near the analog section: the charger at 3 MHz (only while plugged in), the LTC3440 at 1.2 MHz and the LT3462A at 2.7 MHz (both always running). Switching frequencies well above the audio band sit inside the LDOs' effective PSRR range, so conducted noise is manageable. Magnetic near-field coupling and ground-return currents are not filterable and are handled by placement, shielded inductors and return-path control.
- No spread-spectrum modulation on the analog supply path. X5R/X7R DC-bias derating is severe; parts are specified against the derated figure.

4.6 Display integration

- The design integrates a bare 2.4 inch panel with an FPC connector.
- Motivation is thickness, not cost. At 2.4 inch the glass fits inside the battery footprint, so the display costs no additional volume.
- Requires a 0.5 mm or 0.3 mm pitch FPC connector and tail routing, plus a constant-current boost LED backlight driver (typically 3–4 LEDs in series at 9–20 V forward).

- The backlight driver is a further switcher. It sits away from the analog section, uses a shielded inductor, and keeps its PWM dimming frequency and harmonics clear of the audio band.
- Documentation risk: the panel must come from a vendor publishing its FPC pinout and backlight configuration.

5. Target Specifications

Parameter	Target	Comment
Output connector	3.5 mm TRS, single-ended	DC coupled, one jack.
Rated load	Simgot EW300, 28 Ω $\pm 15\%$	16–32 ohm IEMs supported.
Output impedance	< 1 ohm, 20 Hz – 20 kHz	Most important spec for this load.
Max output	0.6 V RMS	≈ 15 mW into 23.8 Ω ; 114.6 dB SPL peak on the EW300.
Output noise	< 3 μ V RMS, A-weighted	Audible hiss floor is the practical limiter.
THD+N	< –105 dB @ 1 kHz, 32 ohm	The DAC is not the limiting element.
Crosstalk	< –80 dB @ 1 kHz	Single-ended shared ground return is the limit.
Formats	FLAC 16/44.1 and 16/48, MP3	24-bit and higher rates are a stretch goal.
Battery life	≥ 7 h continuous playback	Budget predicts 7.5 h with full derating.
Charge time	~ 4.5 –5 h at 500 mA	Including CV taper. Overnight charging assumed.
Enclosure	$\sim 56 \times 75 \times 13.2$ mm	Driven by the 63 x 51 x 6.6 mm cell envelope.

6. Simulation Plan

Scope note. Simulation is applied where it changes design decisions, not as a blanket rule. The power tree's switchers and LDOs qualify: PSRR, output impedance, noise and ripple spectra decide whether the noise-floor target is met. The charger does not: its CC/CV state machine, termination detection, JEITA foldback and PowerPath arbitration are digital behaviours SPICE does not model, and its design work is datasheet arithmetic and thermal layout instead.

6.1 LTspice: buck-boost, inverter, LDOs

- LDO PSRR versus frequency, focused on 100 kHz – 3 MHz where the switchers' energy lands.
- Output noise spectral density and integrated RMS noise over the audio band.
- LDO output impedance versus frequency, which determines rejection of load transients from the DAC and output stage.
- Buck-boost and inverter transient behaviour, ripple spectrum, inductor current at worst-case load.
- Startup, UVLO and load-step response, including the SD card's read-burst step.
- Backlight boost ripple and its coupling path.

6.2 Signal chain

- I/V stage settling, noise contribution, stability against DAC output capacitance.
- MFB filter magnitude, phase, stopband into the MHz region, tolerance sensitivity. Compared against an equivalent Sallen-Key to confirm the stopband argument empirically.
- OPA1622 loop gain and phase margin into capacitive cable loads, with and without output-inclusive compensation.
- Closed-loop output impedance versus frequency, verifying the sub-1-ohm target.
- Gain structure and clipping across the volume range.
- Full-path noise analysis with each contributor identified.

6.3 Hand calculation, not simulation

- Charger inductor ripple and peak current, from the datasheet equations.
- TS network resistor values, from the two simultaneous ratiometric threshold equations.

- Charger junction temperature from theta-JA and dissipation.
- Charge time from capacity, current and CV taper behaviour.

7. Risk Register

Risk	Impact	Mitigation
Battery 28-week lead time	Six-month stall if stock lapses	Purchase two units immediately. The second is needed anyway for discharge characterisation.
No internal NTC in the cell	No hardware charge temperature qualification	Resolved by the BQ24250 TS pin plus an external NTC on the cell body. Moved from firmware responsibility to hardware.
Charge time approaches the safety timer	Charging aborts before completion	Set the 9 h timer via I2C. The 6 h default is not relied on with a ~4.5–5 h charge.
Charger switching noise near the analog chain	Noise floor degradation while plugged in	3 MHz fundamental is well inside the LT3042's PSRR range. Placement, shielded inductor, minimised loop area for the unfilterable magnetic and return-path coupling. The charger is inactive on battery, which is when listening happens.
Charge current at the bottom of range	Reduced margin on the ICHG setting	500 mA is the specified minimum and the datasheet's own design example. The LOW_CHG bit forces 330 mA if less is ever wanted.

Risk	Impact	Mitigation
LT3462A switch current at full-scale output	Negative rail clips at full scale	≈ 0.29 A vs 300 mA minimum at the 98 mA peak; ≈ 0.21 A at loud listening. Confirm against the current-limit vs duty-cycle curve.
OPA1622 is 10-pin VSON, thermal pad at V–	Cannot be hand-soldered; pad is at -3.3 V	Hot air or hotplate with a stencil. Dedicated V– pour and via field, never merged into ground.
DC-coupled output fault	Destroys an IEM	Offset detection with a muting device. Required.
SD read latency causing underrun	Audible dropouts	Ring buffer sized against worst-case latency; minimum occupancy logged.
Voltage-based gauging inaccuracy	Misleading battery indicator	Load compensation, measured lookup table, ADC factory calibration. Li-ion discharge curves are deliberately flat mid-range.
USB MSC and playback contending for the volume	Filesystem corruption	State machine: enumeration stops playback and releases the volume before host mount.
Scope creep	Project never finishes	Section 3 is the contract.

8. Open Questions

- **NTC selection.** R25 and beta determine R2 and R3 from the two ratiometric threshold equations.
- **LT3462A inductor selection.** Shielded, 22 μ H, $I_{\text{sat}} \geq 0.35$ A, with efficiency from LTspice.
- **OPA1622 quiescent current** at 6.6 V total, read from the Quiescent Current vs. Supply Voltage curve, plus its shutdown current for the standby figure.

- **Summing-stage op-amp and gain.** Quiescent current is a selection criterion. Gain worked backward from the 0.6 V RMS output target.
- **Volume control method:** digital attenuation in the DAC, analog potentiometer, or digital potentiometer ahead of the output stage. Affects the noise floor at low listening levels.
- **Output muting and DC-fault protection topology.** Candidate: driving the mute from the LDO PG outputs in hardware.
- **Display panel vendor,** with confirmation that the FPC pinout and backlight configuration are published.
- **Measurement equipment:** a -105 dB target needs an analyser floor below it.

Sections 7 and 8 are open by design.